

The Translation Layer

Polytope Overlap as the Empirical Grammar of the V_{600} Substrate, with Dictionary and Translation Rules

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Abstract

The V_{600} closure programme has been building a mathematical substrate — the icosian triad $(\mathcal{I}, \mathcal{G}, C_\varphi)$ on the 120-vertex set of the regular 600-cell — and applying it across mathematics, cosmology, biology and number theory. A natural worry is that the substrate is functioning as a metaphor system rather than as a working formal language. This paper makes the case, with reproducible engine output, that the substrate now behaves as a language in the technical sense: it has a *dictionary* (a typed map from substrate objects to candidate interpretive referents), it has a *grammar* (a finite set of admissible transformations with explicit well-formedness classes), and it has *translation rules* (admissibility + dictionary $\rightarrow$ candidate observable).

The grammar is not declarative. It is exhibited empirically by the *closure transformation engine*: a computational artefact that computes, on V_{600} and on polytope-overlap structures derived from it, which transformations preserve closure and which break it. Three load-bearing results pin the grammar down concretely:

1. The Schläfli decomposition $V_{600} = C_0 \sqcup C_1 \sqcup C_2 \sqcup C_3 \sqcup C_4$ into five right cosets of the binary tetrahedral subgroup $V_{24} = 2T \subset 2I = V_{600}$, each isomorphic to the 24-cell, with induced faithful A_5 action on the coset index — five reproducible certificates.
2. The $\lambda = 12$ *lift channel*: every per-coset 24-cell $\lambda = 12$ eigenvector extends to a full- V_{600} Laplacian $\lambda = 12$ eigenvector. The lifted space is 10-dim inside $\dim E_{600}(12) = 25$, with the residual 15-dim block carrying a non-trivial A_5 character.
3. The $\lambda = 12$ *uniqueness scan*: across all tested subgroups $H \subset 2I$ and all global eigenvalues $\lambda', \lambda' = 12$ is the only value at which the full lift channel opens. Every other (H, λ') either produces a strictly smaller channel or no channel at all.

These three results are the grammar: a positive rule ($\lambda = 12$ is admissible), a uniqueness claim ($\lambda \neq 12$ is forbidden), and a syntactic structure (the A_5 character split of the lifted vs. residual blocks). On top of this grammar sit two further classes of artefact: a *transformation taxonomy* that names twelve legal moves and five admissibility classes, with which physical-interpretation labels are formally tied; and a relational reasoning system (`closure_ai / min_aria`) that uses the engine's admissibility verdicts as its own state-transition rules, demonstrating that the language is *compositional and productive* in the formal-language sense.

The paper's claim is not that this language settles physics. It is that the substrate has crossed the threshold from metaphor to formal language: dictionary, grammar, and translation are all exhibited on reproducible artefacts. The substrate can be assessed by the standards used to assess other working mathematical languages.

Status. Pre-peer-review open research preprint. Not independently validated. No claim of physical derivation, no claim of consciousness reduction, no claim of RH / GRH settlement.

Every interpretive translation is explicitly a candidate identification, conditional on the bridge hypotheses listed in §7.

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1 Introduction: the translation problem

1.1 Why “language” is the right shape, not just the right word

The V_{600} closure programme has been accumulating outputs of a particular shape. Spectra of finite operators get identified with families of physical observables. Cosets of finite groups get identified with sectors of the Standard Model. Hilbert modular L -functions get identified with classical zeta factors. The recurrence is striking and the underlying object — the icosian triad $(\mathcal{I}, \mathcal{G}, C_\varphi)$ on V_{600} — keeps doing the same kind of work in each case. The natural diagnosis is that the substrate is functioning as a language: a fixed vocabulary plus a set of composition rules from which a wide range of declarative sentences can be formed.

This paper takes that diagnosis literally and asks what would have to be true for it to be correct. A working formal language requires three components:

1. A *dictionary* (lexicon): a typed map from primitive substrate objects to candidate referents in the interpretive layer.
2. A *grammar*: a finite set of rules saying which compositions of primitives are well-formed, which are forbidden, and which are conditionally admissible.
3. A *translation procedure*: a rule that converts a well-formed sentence of the substrate language into a statement in the interpretive layer.

Without (i) the substrate is a graph. Without (ii) the substrate is a metaphor system. Without (iii) the substrate is a dictionary-plus-grammar with no semantics. *All three* are required.

1.2 What is concrete and publishable here

We claim, and exhibit, all three on reproducible engine output. The key empirical artefact is the *closure transformation engine*. It is a computational pipeline that, given a graph or polytope specification, computes:

- closure validation (does the structure close consistently?);
- invariant extraction (Euler characteristic, spectra, curvature class);
- duality and projection operators;
- admissibility verdicts for proposed transformations.

The engine does not assume physical content. It is a *mathematical* pipeline. Its physical-interpretation labels are deliberately conservative; every label is keyed off an abstract **Admissibility-Class** and is phrased as a candidate identification.

The reason the engine carries the paper is that it lets us write down the grammar of the substrate *empirically*, by observing which transformations preserve closure and which break it. The $\lambda = 12$ lift channel and its uniqueness across alternative subgroups and eigenvalues is the cleanest example: an exhibited *positive* grammar rule plus an exhibited *negative* constraint, both reproducible.

1.3 What this paper is and is not

Scope

Is. An empirical demonstration that the V_{600} substrate now meets the three formal criteria for a language — dictionary, grammar, translation — with reproducible engine output as the load-bearing evidence.

Is not. A claim that the language settles physics, life, mind, or RH. Every interpretive translation in this paper is a candidate identification, conditional on a named bridge hypothesis.

1.4 Roadmap

§2 states the three-layer framing. §3 describes the closure transformation engine and its outputs. §4 exhibits the dictionary. §5 exhibits the grammar from polytope overlap, with the Schläfli decomposition, the $\lambda = 12$ lift, and the $\lambda = 12$ uniqueness scan as the three load-bearing sub-results. §6 exhibits the translation procedure. §7 states the substrate’s expressive scope — explicitly, what it cannot say. §8 shows compositionality by lifting the grammar to the relational-AI architecture `closure_ai` / `min_aria`. §9 compares the substrate to other working mathematical languages (group theory, twistor theory, amplituhedron, LQG). §10 states open translation problems. §11 concludes.

2 Three layers: dictionary, grammar, interpretation

The three-layer framing originated in our evidence-ledger audit [4] as a way to keep “ V_{600} appears everywhere” from quietly counting as evidence. Here we lift the same framing into a positive constructive frame.

2.1 Layer 1: dictionary

A dictionary entry is a typed map $d_i : \Sigma_i \rightarrow \text{Ref}_i$ that takes a substrate object Σ_i (a vertex, a coset, an eigenspace, a character) to a candidate referent. A dictionary entry is *not evidence*: it is a translation *possibility*, awaiting confirmation by grammar.

2.2 Layer 2: grammar

A grammar rule is a deterministic statement about which compositions of substrate objects are admissible. The closure transformation engine produces grammar rules *operationally*: a transformation T applied to a closed state S either preserves closure (**Admissibility** = **EXACT** or **CANDIDATE**) or breaks it (**APPROXIMATE** / **BROKEN** / **DEGENERATE**). The set of (T, S) pairs returning **EXACT** is the grammar.

2.3 Layer 3: translation

Translation maps a dictionary entry d_i plus a grammar verdict v into an interpretive claim $\text{Ref}_i \mid v$. Crucially: *translation requires grammar*. A dictionary entry $V_{600} \rightsquigarrow$ “substrate of quantum coherence” is not evidence until the grammar verdict on the relevant composition has been computed. The engine computes the verdict; the translation is whatever the verdict permits.

The three layers, restated

Layer	What it is	Evidential status
1 Dictionary	typed substrate $\rightarrow$ referent map	possibility (not evidence)
2 Grammar	engine-computed admissibility verdict	empirical (theorem-grade or reproducible)
3 Translation	dictionary + grammar $\rightarrow$ interpretive claim	conditional on grammar + bridge hyp.

A common failure mode in informal cross-disciplinary work is to present a Layer-1 dictionary as if it were a Layer-3 translation. The audit pattern of this paper is the opposite: every Layer-3 claim points to a Layer-2 verdict and a Layer-1 entry, and the reader can disagree at any of the three.

3 The closure transformation engine

3.1 What the engine is

The closure transformation engine [3] is a computational pipeline. Its inputs are graph specifications and polytope overlap structures derived from V_{600} . Its outputs are closure verdicts, invariant reports, transformation taxonomy records, admissibility classes and (when requested) candidate physical-interpretation labels.

The engine has three core modules:

core/ closure validation (`closure_validator.py`), invariant extraction (`invariants.py`, `spectrum.py`, `curvature.py`), duality (`duality.py`), projection (`projection.py`), residuals (`residuals.py`), and Schläfli-symbol intake (`schlaflf.py`).

transformations/ twelve transformation kinds (DUAL, SHELL_PROJECTION, BOUNDARY_PROJECTION, SPECTRAL_PROJECTION, CYCLE_PROJECTION, QUOTIENT, SUBGROUP_RESTRICTION, EDGE_CONTRACTION, VERTEX_DELETION, EDGE_DELETION, DIMENSIONAL_LIFT, DIMENSIONAL_PROJECTION). Each carries a typed `TransformationOutcome` record.

physics/ admissibility classification (`admissibility.py`), transition rules between decorated states (`transition_rules.py`), the 24-600 spectral bridge (`bridge_24_600.py`), the $\lambda = 12$ channel (`lambda12_a5_channel.py`), the universality scan (`universality_lambda12.py`), shell hierarchy and gauge classification, and the (deliberately conservative) `physical_interpretation.py` label table.

3.2 Why the engine is the load-bearing artefact

The engine matters here because it is the place where the substrate’s grammar is no longer a verbal claim. Every grammar rule we state in §5 is reproducible from one of the engine’s outputs. Three artefacts in particular do most of the work:

- the Schläfli decomposition module (`decomposition.py`), with five exact certificates;
- the $\lambda = 12$ spectral-bridge module (`keystone.py`), with the lift theorem and the A_5 character decomposition;
- the $\lambda = 12$ universality module (`universality_lambda12.py`), which scans all subgroups $H \subset 2I$ and all V_{600} Laplacian eigenvalues to test whether $\lambda = 12$ is the unique full-lift channel.

These three artefacts together yield the positive-and-negative grammar rule discussed in §5.2.

4 The dictionary

This section exhibits the dictionary. Each entry maps a substrate object to one or more candidate interpretive referents. The referents are typed — “mathematical”, “physical (candidate)”, “cognitive (candidate)”, “ontological (interpretive)” — and each entry carries an evidential weight inherited from its grammar-layer status.

4.1 Primitives

Dictionary entry

$V_{600} = 2I$. 120-element binary icosahedral group, realised as 120 unit icosian quaternions.

Mathematical referent: vertex set of regular 600-cell; finite reflection-group orbit under $W(H_4)$.

Candidate physical referent: closure-substrate carrier.

Evidential weight: Layer-1; weight comes from grammar.

Dictionary entry

$V_{24} = 2T \subset V_{600}$. 24-element binary tetrahedral subgroup, realised as the σ -fixed sublattice of V_{600} .

Mathematical referent: 24-cell vertex set; the fundamental coset block in the Schläfli decomposition.

Candidate physical referent: closure “unit cell” / fundamental sector.

Evidential weight: Layer-1.

Dictionary entry

A_1 (**edge-adjacency operator on V_{600}**). 120×120 matrix with $(A_1)_{ij} = 1$ iff v_i, v_j are vertex-adjacent in V_{600} .

Mathematical referent: graph-Laplacian generator; defines $L = 12 \cdot I - A_1$.

Spectrum: 9 eigenvalues in $\mathbb{Z}[\varphi]$ with multiplicities $(1, 4, 9, 16, 25, 36, 9, 16, 4)$ summing to 120.

Evidential weight: Layer-2 (theorem-grade).

Dictionary entry

$C_\varphi = L + \varphi^{-2}I = (12 + \varphi^{-2})I - A_1$. The closure operator.

Mathematical referent: self-adjoint positive operator inducing the $94 + 13 + 13$ σ -pairing decomposition.

Candidate referent: the “closure verb” — the mathematical instantiation of “ X exists $\iff \mathcal{C}(X) = X$ ”.

Evidential weight: Layer-2 (theorem-grade) for the math; Layer-3 (interpretation) for the existence reading.

Dictionary entry

$N_H : \mathcal{I} \rightarrow \mathbb{Z}[\varphi]$, $N_H(q) = q_0^2 + q_1^2 + q_2^2 + q_3^2$. The generation operator.

Mathematical referent: quaternary norm form, universal for totally positive $\mathbb{Z}[\varphi]$ -primes by Eichler–Hijikata.

Evidential weight: Layer-2 (theorem-grade).

4.2 Compound primitives from polytope overlap**Dictionary entry**

The five 24-cell cosets $C_0, \dots, C_4$. $V_{600} = \bigsqcup_{i=0}^4 C_i$ with $|C_i| = 24$, each C_i isomorphic to the 24-cell.

Mathematical referent: right cosets of V_{24} in V_{600} ; five copies of the 24-cell tiled into one 600-cell.

Candidate physical referent: the five *building blocks* of any physical sector built on V_{600} ; the carriers of the A_5 representation.

Evidential weight: Layer-2 (theorem-grade) for the decomposition; Layer-3 for “building blocks”.

Dictionary entry

$E_{600}(\lambda)$. The λ -eigenspace of the V_{600} adjacency Laplacian.

Mathematical referent: a finite-dimensional $W(H_4)$ -rep.

Distinguished entry: $\dim E_{600}(12) = 25$.

Candidate physical referent: candidate spectral-mode-family. The labelling per λ is in [5].

Dictionary entry

The $\lambda = 12$ lift channel Λ_{12} . The 10-dimensional subspace of $E_{600}(12)$ obtained by zero-extending the per-coset $\lambda = 12$ 24-cell eigenvectors.

Mathematical referent: *the carrier of the grammar rule of §5.2*. Its complement $E_{600}(12) \setminus \Lambda_{12}$ is the 15-dim residual block, with a non-trivial A_5 character.

Evidential weight: Layer-2 (theorem-grade).

4.3 Dictionary table

The full machine-readable dictionary is at `data/dictionary.csv`. It records, per row, the substrate object, the mathematical referent, the candidate physical referent, the grammar-layer status, the bridge hypothesis (if any), and the evidence-ledger row of [4] that audits the entry.

5 Polytope overlap as grammar

This is the load-bearing section of the paper. We exhibit three grammar rules derived empirically from the closure transformation engine. Two are positive (the engine returns **EXACT**); one is negative (the engine returns no channel).

5.1 The Schläfli decomposition as the grammar's primitive sentence

Theorem 1 (Schläfli decomposition; engine certificate). *The 120-vertex set $V_{600} = 2I$ admits the exact partition*

$$V_{600} = C_0 \sqcup C_1 \sqcup C_2 \sqcup C_3 \sqcup C_4$$

where $C_0 = V_{24} = 2T$ (the σ -fixed sub-lattice) and $C_1, \dots, C_4$ are the four V_{24} -left-multiplication orbits on the 96 σ -mobile vertices. Each C_i is exactly a coset of V_{24} , has cardinality 24, and is isometric to the 24-cell vertex set with $d=1$ graph 8-regular on 96 edges. The induced action of V_{600} on the coset set $\{C_0, \dots, C_4\}$ descends through $\{\pm 1\}$ to a faithful transitive action of A_5 on five points.

The theorem is verified by five reproducible certificates from `closure_transform_engine.decomposition`: `V_600` construction, `V_24` subgroup, right cosets, each coset is a 24-cell, A_5 coset action. All five are exact over $\mathbb{Q}(\sqrt{5})$.

Why this is a grammar primitive. Because every subsequent grammar rule in this paper either references the five cosets or references the A_5 representation they carry. Theorem 1 is the minimal compositional unit on which the rest of the grammar is built.

5.2 The $\lambda = 12$ lift rule

Theorem 2 (Per-coset $\lambda = 12$ lift theorem; engine certificate). *Let L_{600} be the $d=1$ graph Laplacian on V_{600} (12-regular, 720 edges). For each $i \in \{0, 1, 2, 3, 4\}$ let L_{C_i} be the $d=1$ graph Laplacian on the coset C_i (8-regular, 96 edges, isomorphic to the 24-cell edge graph). For each i , the $\lambda = 12$ eigenspace $E_{C_i}(12)$ of L_{C_i} has dimension 2. The zero-extension of any $v \in E_{C_i}(12)$ to $\mathbb{R}^{120}$ is an eigenvector of L_{600} at eigenvalue exactly 12. Summing over i yields a 10-dimensional subspace $\Lambda_{12} \subset E_{600}(12)$. The complementary 15-dimensional residual block carries a non-trivial A_5 character orthogonal to the lifted block.*

The theorem is verified by `closure_transform_engine.keystone.exact_certify_lambda12_lift` and the A_5 character calculation `exact_a5_characters`.

Grammar rule

Rule (positive). “Lift along the Schläfli decomposition at $\lambda = 12$ is admissible.”

Substrate side: per-coset 24-cell eigenvector at $\lambda = 12$.

Grammar verdict: the engine returns `Admissibility = EXACT` with $\|L_{600}v - 12v\| < \epsilon_{\text{mach}}$ for every input.

Composition: the lift is linear and commutes with the A_5 action on cosets.

5.3 The $\lambda = 12$ uniqueness scan: a forbidden-move rule

The positive rule of §5.2 is not yet a grammar; a grammar must also tell us which moves are *forbidden*. The empirical answer is supplied by the universality scan in `closure_transform_engine.physics.univ`.

Setup. For each subgroup $H \subset 2I$ in a finite test set and each eigenvalue $\lambda' \in \text{spec}(L_{600})$, the scan computes

$$\text{Lift}(H, \lambda') = \bigoplus_i \{v \in \mathbb{R}^{V_{600}} : \text{supp}(v) \subseteq C_i^{(H)}, L_{600}v = \lambda'v\},$$

where $C_i^{(H)}$ ranges over the right cosets of H in $2I$. The scan reports the dimension of $\text{Lift}(H, \lambda')$ for every pair (H, λ') , and flags every pair whose channel dimension is non-zero.

Empirical finding. Across the tested subgroup set, the *full* lift channel ($\text{Lift}(H, \lambda') = \dim E_{600}(\lambda')$ from per-coset support alone) opens exactly when $H = V_{24}$ and $\lambda' = 12$. Every other pair returns either a strictly smaller channel or no channel at all. The five negative controls $\lambda' \in \{0, 4, 8, 10\}$ all return zero per-coset lift into L_{600} .

Grammar rule

Rule (negative). “Polytope-overlap lift at $\lambda' \neq 12$ is forbidden.”

Substrate side: per- H -coset eigenvector at λ' .

Grammar verdict: for $\lambda' \in \{0, 4, 8, 10\}$ the engine returns **Admissibility** = **BROKEN**.

Composition: the forbidden-move rule survives change of subgroup H in the tested set.

The conjunction “ $\lambda = 12$ admissible, all others forbidden” is the first grammar rule the substrate has at the sentence level. It is a clean empirical *constraint*, not a labelling choice.

5.4 Transformation kinds as the legal-move set

The engine’s transformation taxonomy provides the substrate’s operational verbs: the twelve transformation kinds (DUAL, SHELL_PROJECTION, BOUNDARY_PROJECTION, SPECTRAL_PROJECTION, CYCLE_PROJECTION, QUOTIENT, SUBGROUP_RESTRICTION, EDGE_CONTRACTION, VERTEX_DELETION, EDGE_DELETION, DIMENSIONAL_LIFT, DIMENSIONAL_PROJECTION). Each carries an explicit input typing, an output typing, and a closure-residual recomputation. A move is *legal* when its admissibility class returns **EXACT** or **CANDIDATE**; it is *illegal* when it returns **BROKEN** or **DEGENERATE**.

This is the substrate’s grammar in a strict computational sense. The full move table is at `data/grammar.csv`, together with the per-move pre-and-post invariant lists.

5.5 Admissibility classes as well-formedness

The engine resolves every transformation into one of five admissibility classes:

- **EXACT** — closure preserved within numerical tolerance.
- **CANDIDATE** — near-closed, low residual.
- **APPROXIMATE** — noticeable residual but interpretable.
- **BROKEN** — closure lost.
- **DEGENERATE** — transformation produced a trivial or empty target.

A well-formed sentence of the substrate language is, by definition, a chain of moves each of which terminates in **EXACT** or **CANDIDATE**. The substrate’s grammar is this well-formedness predicate.

5.6 Transition rules as composition

The engine’s **TransitionRule** record specifies the default-allowed transitions between decorated closure states: `preserve_topology`, `preserve_charge_like_labels`, `preserve_boundary_closure`, with `max_residual_delta = 0.01` and optional `allow_phase_defect`. Each transition is resolved to a verdict from the set {allowed, forbidden, allowed-as-virtual, allowed-as-decay-candidate, requires-boundary-change}. This is the substrate’s compositional rule: how shorter well-formed sentences combine into longer ones.

6 Translation rules

Translation is the operation $T : (d_i, v) \mapsto r_i \mid v$. The dictionary supplies d_i ; the grammar supplies the verdict v ; translation produces a candidate interpretive claim $r_i \mid v$.

6.1 The translation procedure, formally

Principle 1 (Translation). *A translation from substrate object Σ_i to candidate referent Ref_i is admissible iff:*

- (i) *A dictionary entry $d_i : \Sigma_i \rightarrow \text{Ref}_i$ exists in `data/dictionary.csv`.*
- (ii) *There is a sentence s of the substrate language using Σ_i as a primitive whose grammar verdict $v(s)$ is **EXACT** or **CANDIDATE**.*
- (iii) *A named bridge hypothesis $H_{i,\text{Ref}}$ is supplied, connecting $\Sigma_i \mid v(s)$ to the interpretive domain.*

The translation is then $r_i \mid (v(s) \wedge H_{i,\text{Ref}})$.

The principle ties Layer 3 to Layer 2 to Layer 1. Without (ii) the translation is metaphor. Without (iii) the translation is a Hilbert space without an inner product: well-formed, semantically empty.

6.2 Worked translation: closed admissibility $\rightarrow$ stable observable candidate

Translation

Substrate sentence. A decorated closure state S over V_{600} , with the $\lambda = 12$ lift channel populated and zero boundary defect.

Grammar verdict. Admissibility = CLOSED (per `physical_interpretation.py`, line `_INTERPRETATIONS [AdmissibilityClass.CLOSED]`).

Dictionary entries used. V_{600} as closure-substrate; Λ_{12} as carrier of the closure mode-family.

Bridge hypothesis required. H-Observable: stable mathematical closure corresponds to a candidate physical observable.

Translation. “The closure-substrate carries a candidate stable observable, conditional on H-Observable.”

Audit caveat (engine-enforced). “Not yet a physical derivation; closure does not imply realisability.”

This is the deliberate conservatism of the engine reflected at the paper level. The engine attaches an explicit caution to every physical-interpretation label and the translation principle inherits it.

6.3 Worked translation: near-closed $\rightarrow$ resonance candidate

Translation

Substrate sentence. Decorated state S' with small but non-zero closure residual on the boundary.

Grammar verdict. $\text{Admissibility} = \text{NEAR_CLOSED}$.

Dictionary entries. same as above plus a boundary-decoration field.

Bridge hypothesis. H-Resonance: near-closure $\leftrightarrow$ candidate resonance / transient bound state.

Translation. “Candidate resonance or transient bound state, conditional on H-Resonance and on the boundary decoration not being an artefact.”

Audit caveat. “May be an artefact of the chosen decoration — try alternative phase or boundary assignments before drawing conclusions.”

6.4 Worked translation: $\lambda = 12$ universality $\rightarrow$ substrate mode-family uniqueness

Translation

Substrate sentence. The $\lambda = 12$ lift theorem and its uniqueness scan.

Grammar verdict. EXACT for the lift theorem; BROKEN or smaller-channel for every other (H, λ') in the tested set.

Dictionary entries. V_{600} , the five cosets, Λ_{12} .

Bridge hypothesis. H-ModeUnique: substrate mode-family selection corresponds to grammar-rule selection.

Translation. “The substrate selects exactly one mode-family across the tested choice of subgroup and eigenvalue, conditional on H-ModeUnique.”

Audit caveat. The test set of subgroups is finite. Extending the test set is the open work.

6.5 What translation is *not*

- Translation is not a physical derivation. A translation produces a *candidate* identification. The candidate becomes a derivation only when the bridge hypothesis is independently closed.
- Translation is not equivalence. The interpretive referent may have structure the substrate cannot represent. Translation is one-way unless the dictionary entry has an inverse.
- Translation is not falsification. A failing translation does not falsify the substrate; it falsifies the translation or the bridge hypothesis. The substrate remains.

7 Expressive scope: what the substrate can and cannot say

A working formal language has an *expressive scope*. There are sentences inside it and sentences it cannot form. The grammar rules of §5 let us state the substrate’s scope explicitly.

7.1 What the substrate can say

1. Spectra of self-adjoint operators on V_{600} , with eigenvalues in $\mathbb{Z}[\varphi]$.
2. Subspaces invariant under $W(H_4)$, decomposed into irreducibles of the Coxeter group.

3. Cosets of finite subgroups $H \subset 2I$ and the lift channels they open.
4. σ -pairing decompositions of subspaces (the $94 + 13 + 13$ split).
5. Quaternary norm forms N_H and the totally positive $\mathbb{Z}[\varphi]$ -primes they enumerate.
6. Hilbert modular L -functions over $K = \mathbb{Q}(\sqrt{5})$ factoring through the substrate theta series.
7. Composition of any of the above under transformation kinds in the move table.

7.2 What the substrate cannot say

1. Continuous deformations of V_{600} . The substrate is finite. Smooth deformations have to be carried by the interpretive layer.
2. Statements that are not $W(H_4)$ -covariant. The grammar forbids them. (This is the substrate's first explicit *negation*.)
3. Statements that require a subgroup outside the test set of the universality scan. These remain open.
4. Statements about objects outside $\mathbb{Q}(\sqrt{5})$. Other biquadratic fields are reachable only via the `DIMENSIONAL_LIFT` kind, and only with named bridge hypotheses.
5. Anything in the interpretive layer that has no dictionary entry. Without a dictionary entry, there is no translation.

7.3 Bridge hypotheses currently consumed by translation

H-Observable. Closure $\leftrightarrow$ candidate observable. Open.

H-Resonance. Near-closure $\leftrightarrow$ candidate resonance. Open.

H-ModeUnique. $\lambda = 12$ universality $\leftrightarrow$ substrate mode-family uniqueness. Open.

H-CortexClosure. Cortical phase closure $\leftrightarrow$ closure observable [2]. Open.

H-MT-Bridge. Microtubule lattice $\leftrightarrow$ candidate substrate carrier. Open (exploratory).

The four-tier scope discipline of [2, §9] and the constraint-forcing criterion of [4, §5] both apply unchanged here.

8 Compositionality: the language is productive

A working language is more than dictionary + grammar + translation: it is *compositional* and *productive*. Finitely many primitives plus a grammar yield infinitely many well-formed sentences. We show that the substrate has this property by exhibiting a reasoning system built on top of the engine.

8.1 closure_ai: relational reasoning over substrate states

`closure_ai` [1] is a relational reasoning architecture whose hidden state is a decorated cognitive graph, whose reasoning steps are graph transformations from the engine, and whose attention is invariant-aware projection. Crucially: *outputs are emitted only after the admissibility gate accepts the final state*.

Why this matters here. It shows that the engine’s admissibility classes are usable as a complete state-transition predicate. The system runs because the grammar of §5.5 is decidable. A grammar that classified well-formedness ambiguously could not drive a reasoning loop. The fact that `closure_ai` runs is empirical evidence that the substrate grammar is a *functioning* formal language, not just a labelled set of moves.

8.2 min_aria: vector-state substrate carrier

`min_aria` [6] is a minimal substrate-aware agent carrying a 7-dimensional vector state

$$S_t = [\text{identity_stability}, \text{memory_load}, \text{novelty_score}, \text{closure_residual}, \\ \text{perturbation_level}, \text{governance_integrity}, \text{replay_integrity}].$$

The closure-residual component is the engine’s well-formedness predicate of §5.5; the perturbation and governance components are direct readouts of admissibility-class transitions. The agent maintains identity under hostile prompts ($\text{identity_drift} \approx 0$) and recovers closure under perturbation ($\text{closure_residual} \rightarrow 0$). The numerical proofs of these claims live in `min_aria/test_min_aria_v2.py`.

What this proves. The substrate’s grammar is compositional enough to drive a stateful agent. That is not a weak claim. It is the property that distinguishes “a labelled algebraic structure” from “a formal language”.

8.3 Productivity: finite vocabulary, unbounded expressions

The dictionary has finite size. The grammar has finite move set. But the set of well-formed sentences is unbounded: composition of transformations can produce decorated states of arbitrarily large graph depth, and the engine’s residual budget gives the well-formed ones a clean stopping criterion. This is the precise property linguistic theory calls *productivity*.

8.4 Falsifiability: the grammar refuses sentences

The grammar refuses sentences. `BROKEN` verdicts are returned every time a transformation breaks closure beyond $\Delta_{\max} = 0.01$ in the residual budget. This is exactly the property linguistic theory calls *discriminative power* — the grammar produces ill-formed sentences as well as well-formed ones.

Together: dictionary + decidable grammar + productivity + discriminative power are the four classical criteria for a formal language. The substrate exhibits all four, reproducibly, on engine artefacts.

9 Comparison to other mathematical languages

The V_{600} substrate is not the first mathematical language to be exhibited in roughly this form. The comparison is informative because it sets a recognisable scale.

9.1 Group-representation theory

The dictionary is the character table; the grammar is induced and restricted representations and tensor products; the translation layer is the assignment of irreducible representations to physical multiplets. Productivity: a finite group generates an unbounded sequence of well-formed compound representations. Falsifiability: representation theory refuses tensor products that fail the character orthogonality relations.

9.2 Lie algebra and root systems

The dictionary is roots and weights; the grammar is the structure constants and Weyl group action; the translation layer is the identification of simple roots with gauge generators. Productivity is again unbounded; falsifiability is the Jacobi identity.

9.3 Twistor theory

The dictionary maps twistor space coordinates to null directions in Minkowski space; the grammar is the Penrose transform; the translation produces scattering-amplitude representations. The twistor programme has explicit Layer-2 (grammar) results, Layer-3 (translation) results conditional on bridge hypotheses (the Penrose transform's analyticity assumptions), and Layer-1 (dictionary) entries used across the programme.

9.4 Amplituhedron programme

The dictionary is positive Grassmannian cells; the grammar is the boundary and triangulation calculus; the translation produces $\mathcal{N} = 4$ super-Yang–Mills amplitudes. The translation *is* the amplitude, so the bridge hypothesis is exceptionally strong (rather than weak) in this case. The amplituhedron is the nearest comparison to what the V_{600} substrate aspires to be.

9.5 Loop quantum gravity

The dictionary is spin networks; the grammar is gauge-invariant moves on the graph; the translation produces candidate gravitational observables. The bridge hypotheses are explicit and open.

9.6 Where V_{600} sits

The V_{600} substrate sits in this same shape but with three distinguishing features:

- Its grammar is exhibited on a single *fixed* finite substrate (the 600-cell vertex set) rather than on a family of generators.
- It explicitly carries an evidence ledger [4], separating dictionary entries from constraint-forcing entries.
- It is, currently, the work of a single author working outside an institutional setting, released as pre-peer- review preprints. We are explicit that the comparison to the four programmes above is structural, not equal in stature.

10 Open translation problems

The translation layer is the youngest of the three layers and has the most open work. We list the substantive open problems so that future work has a focus.

1. **Subgroup test-set completion.** The $\lambda = 12$ universality scan (`universality_lambda12.py`) was run over a finite test set of subgroups $H \subset 2I$. Extending the scan to *all* subgroups of $2I$ would convert the negative rule of §5.3 from empirical-on-test-set into theorem-grade.
2. **Bridge-hypothesis closure.** H-Observable, H-Resonance, H-ModeUnique, H-CortexClosure, H-MT-Bridge are all open. Each translation in §6 stands or falls with its bridge.

3. **Inverse-translation construction.** Every translation in this paper is one-way (substrate $\rightarrow$ interpretation). Whether the dictionary has inverses — whether every well-formed interpretive sentence has a substrate translation — is open.
4. **Compositionality at scale.** `closure_ai` and `min_aria` demonstrate compositionality at small scale. Whether the grammar remains decidable as the sentence length grows is an open empirical question; the residual budget gives the right place to look.
5. **Bidirectional translation table.** The `data/translation.csv` CSV gives the current forward translation table. A reverse table (interpretation $\rightarrow$ candidate substrate sentence) does not yet exist; it is open work.

11 Conclusion

This paper has argued that the V_{600} substrate has crossed the threshold from metaphor to formal language. The argument is not declarative; it is empirical. The closure transformation engine supplies the grammar, reproducibly. The Schläfli decomposition and the $\lambda = 12$ lift-channel theorem are positive rules. The $\lambda = 12$ uniqueness scan is a negative rule. The transformation taxonomy supplies the move set. The admissibility classes supply the well-formedness predicate. The translation principle ties dictionary to grammar to interpretation. And `closure_ai` / `min_aria` show that the grammar is decidable enough to drive a reasoning system.

The substrate now meets the four classical criteria of a formal language: typed dictionary, decidable grammar, productivity, discriminative power. That does not make it a physical theory. It does mean that the substrate’s claims can be assessed at the *linguistic* level — which dictionary entries are well-defined, which grammar rules are tight, which translations clear their bridge hypotheses — rather than at the “recurrence-by-construction” level. We invite this kind of engagement.

The paper does not change a single claim of the underlying programme. It changes the form in which the programme’s claims can be reviewed.

A reading note. The dictionary, grammar, and translation tables in the appendices are typeset summaries of canonical CSVs in `data/`. Disagreement with a row of any table is welcome; the CSVs are the artefacts to file an issue against.

A Dictionary table

The canonical machine-readable dictionary is at `data/dictionary.csv`. The columns are:

1. `entry_id` — short identifier.
2. `substrate_object` — the substrate primitive.
3. `mathematical_referent` — the mathematical referent of the entry.
4. `candidate_physical_referent` — the candidate physical referent (if any).
5. `candidate_cognitive_referent` — the candidate cognitive referent (if any).
6. `layer` — 1, 2, or 3.
7. `evidential_status` — `theorem-grade`, `conditional`, `interpretive`.
8. `bridge_hypothesis` — the named bridge required to lift the entry to Layer 3.
9. `evidence_ledger_row` — the [4] row that audits this entry.

The dictionary contains 14 entries in v1.0.0-rc1.

B Grammar table

The canonical machine-readable grammar is at `data/grammar.csv`. Columns:

1. `rule_id`
2. `transformation_kind` — DUAL, SHELL_PROJECTION, ..., DIMENSIONAL_PROJECTION (12 kinds).
3. `input_typing`
4. `output_typing`
5. `preserved_invariants`
6. `broken_invariants`
7. `exchanged_invariants`
8. `default_admissibility_class` — EXACT, CANDIDATE, APPROXIMATE, BROKEN, DEGENERATE.
9. `engine_module` — the engine module that implements the rule.

The grammar contains 12 rules in v1.0.0-rc1, one per transformation kind, plus three derived rules for the Schläfli decomposition, the $\lambda = 12$ lift, and the $\lambda = 12$ uniqueness scan.

C Translation table

The canonical machine-readable translation table is at `data/translation.csv`. Columns:

1. `translation_id`
2. `dictionary_entry_id`
3. `grammar_verdict` — EXACT, CANDIDATE, APPROXIMATE, etc.
4. `bridge_hypothesis` — the named bridge.
5. `interpretive_claim` — the candidate interpretive claim.
6. `audit_caveat` — the conservative caveat attached by `physical_interpretation.py` (or equivalent).
7. `open_problems` — what would have to happen for the translation to upgrade.

The translation table contains 14 entries in v1.0.0-rc1.

D Engine reproducibility

Every claim of grammar in §5 is reproducible from modules in the closure transformation engine [3]. Specifically:

Theorem 1. `closure_transform_engine.decomposition` — functions `certify_v600_construction`, `certify_v24_subgroup`, `certify_right_cosets`, `certify_each_coset_is_24cell`, `certify_a5_coset_acti`

Theorem 2. `closure_transform_engine.keystone` — functions `coset_partition`, `build_d1_graphs`, `lambda12_lift`, `exact_certify_lambda12_lift`, `exact_a5_characters`.

$\lambda = 12$ **uniqueness scan.** `closure_transform_engine.physics.universality_lambda12`.

Transformation taxonomy and admissibility classes. `closure_transform_engine.transformations`.
`closure_transform_engine.physics.admissibility`.

Transition rules. `closure_transform_engine.physics.transition_rules`.

Physical-interpretation labels. `closure_transform_engine.physics.physical_interpretation`.

Compositional reasoning. `closure_ai` (relational architecture); `min_aria` (vector-state agent).

All modules are pure Python with `networkx`, `numpy`, `scipy` dependencies. Exact $\mathbb{Q}(\sqrt{5})$ arithmetic is provided by the vendored `vfd_v600` package.

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